PubMed RAG:
AI-Powered Medical Research Retrieval & Analysis

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Problem Statement

PubMed at Scale

36M+ articles, with **1M+ added per year**— the volume of medical literature is overwhelming for any researcher.

Researcher Reality

- Keyword search misses semantically related papers
- Quality assessment (methodology, statistics, clinical applicability) is manual and time-consuming
- Modern workflows involve multimodal inputs (voice notes, images, documents)

Concrete example: "Latest treatments for early-stage pancreatic cancer" — what happens with keyword search?

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Our Approach

Three pillars form the foundation of this system:

1

Semantic + Hybrid Search

Understand **meaning**, not just keywords — surface papers that matter even when terminology differs.

2

Multi-Agent Analysis

8 specialized agents evaluate papers from different angles — methodology, statistics, clinical applicability, and more.

3

Safety

2-layer guardrails + MeSH terminology validation ensure outputs are grounded, accurate, and medically responsible.

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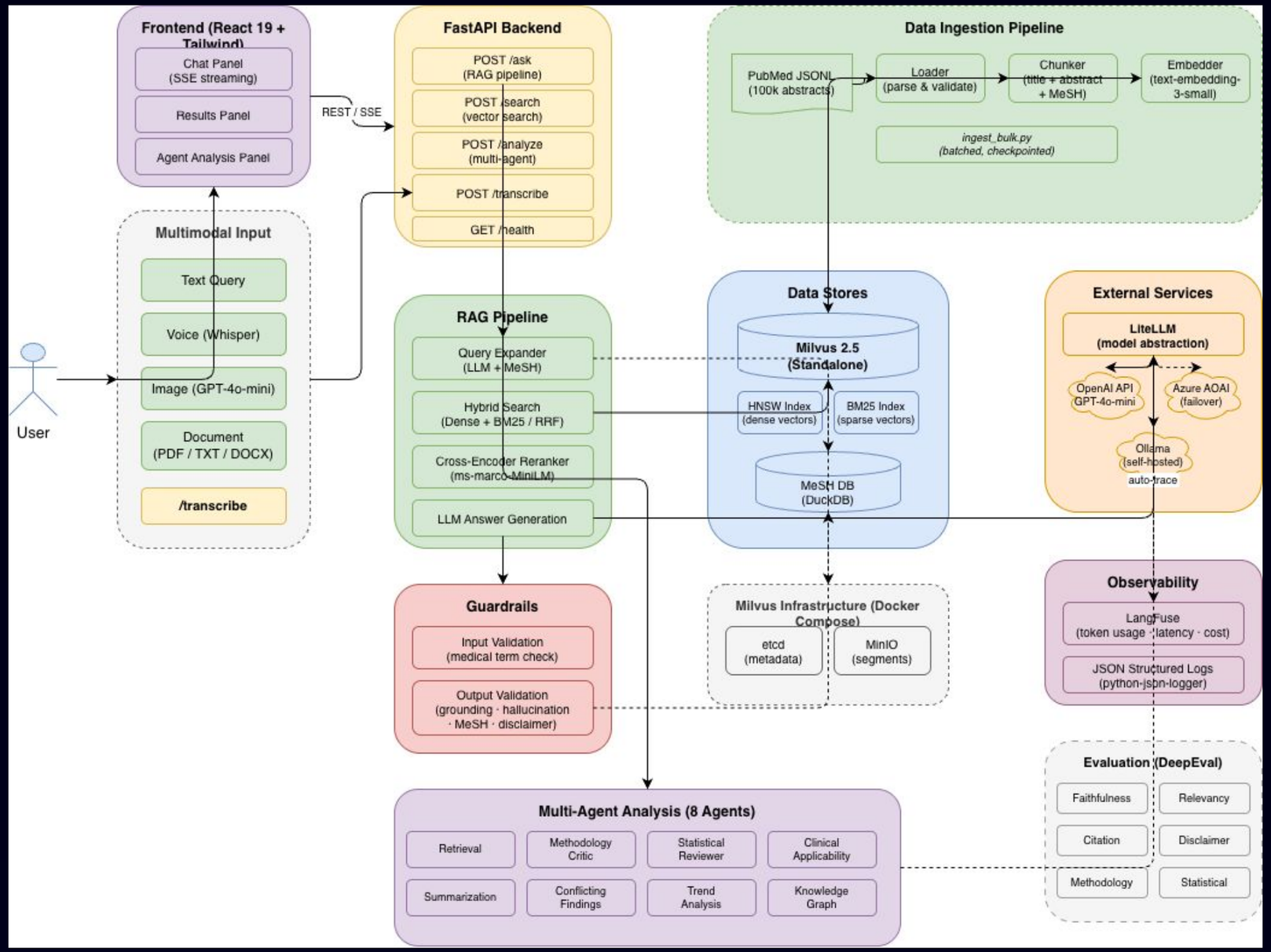
Architecture Overview



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Design Decision — Hybrid Search

The Problem

Dense-only misses exact medical terms.

BM25-only misses semantic similarity.

Neither alone is sufficient for medical literature retrieval.

The Solution

Hybrid Search

Milvus 2.5 native **BM25 + Dense + RRF fusion** — combining the strengths of both approaches.

Query Expansion

1. LLM extracts medical keywords from natural language
2. DuckDB MeSH hierarchy lookup (descriptor → synonyms → child terms)
3. Expanded query sent to hybrid search

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Design Decision — 8 Agents + Protocol DI

Each agent is specialized to evaluate a different dimension of a research paper:

Agent Role	Responsibility
Retrieval Relevance	Assessment of paper relevance to the query
Methodology Critic	Study design & bias evaluation
Statistical Reviewer	Statistical validity & significance
Clinical Applicability	Real-world medical relevance
Summarization	Cross-study synthesis
Conflicting Findings	Inconsistency detection
Trend Analysis	Emerging treatment patterns
Knowledge Graph	Disease-treatment-outcome mapping

Architecture

`BaseAgent Protocol` → all agents share the same interface

Unique Point

Agents reused as **DeepEval custom metrics**: `MethodologyQualityMetric`, `StatisticalValidityMetric`, `ClinicalRelevanceMetric`

Protocol-based DI Throughout

- `SearchClient`
- `BaseReranker`
- `GuardrailClient`
- `BaseAgent`

Enables: easy testing (mock injection), Monolith → Microservice switch

Reference: **ADR-0003**, **ADR-0005**

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Design Decision — Guardrails + Prompt Management

Input Guardrail

LLM-as-judge medical relevance classification

Soft warning, non-blocking — users are informed without being blocked from proceeding.

Output Guardrail — 4 Checks

1. Citation grounding verification
2. Hallucination detection
3. MeSH terminology validation
4. Medical disclaimer auto-attachment

Reference: **ADR-0008**

Prompt Externalization

16 YAML templates covering all prompt surfaces:

Agents ×9

One template per specialized agent

Guardrails ×2

Input and output guardrail prompts

Retrieval ×2

Query expansion and reranking prompts

RAG ×1 + Transcribe ×2

Core RAG chain and multimodal transcription

Change prompts **without code changes** — version-controllable and reviewable by non-engineers.

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Evaluation & Quality

DeepEval — 8 Metrics

Standard:

- Faithfulness
- Answer Relevancy
- Contextual Relevancy

Custom:

- Citation Presence
- Medical Disclaimer
- Methodology Quality
- Statistical Validity
- Clinical Relevance

Testing

Unit tests: 38 files covering all modules

Integration tests: Milvus connection

Eval tests: End-to-end RAG quality

Documentation

16 ADRs (Architecture Decision Records)

- **5 design specs**

Observability

LangFuse integration for token tracking, cost analysis, latency monitoring

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Scope Decisions & Future Work

Scope Decisions

- Full-text retrieval handoff → abstract-level search is the validated core
- Full PubMed corpus (36M+) → 100k stratified sample sufficient for PoC (ADR-0002)

Future Work

- Full-text article retrieval via PubMed Central API
- Scale to full corpus (pipeline is designed for it — change n_max config)
- Circuit breaker + local LLM fallback for offline resilience

Message: Scope was bounded intentionally; architecture supports growth

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Summary

Built: AI-powered medical research retrieval & analysis system

Technical Highlights

1 **Hybrid Search**

Milvus 2.5 native **BM25 + Dense + RRF** — best of both retrieval worlds

2 **8 Specialized Agents**

Reused as **evaluation metrics** — a unique dual-purpose architecture

3 **2-Layer Guardrails**

For **medical safety** — input classification + 4-check output validation

Infrastructure

Docker Compose — 5 services

Architecture

Protocol-based DI throughout

Prompts

16 externalized YAML templates

Thank you → Q&A

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GitHub link: https://github.com/o13o/pubmed-rag

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